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Aim

To study and implement sequence data types in Python such as Strings, Lists, Arrays, Tuples, Dict Sets, and Range along with their operations.

Assignment -1

Theory

1. What are sequence data types in Python?

Sequence data types are used to store multiple values in an ordered manner.

Types of Sequence Data Types:

- String
- List
- Tuple
- Range

Indexing:

- Index starts from 0
- Used to access elements

Example: `s = "Python"` `s[0] → P`

Slicing:

- Used to get part of sequence

Example: `s[1:4] → yth`

2. Difference between List and Tuple

Feature	List	Tuple
Mutable	Yes	No
Syntax	<code>[]</code>	<code>()</code>
Speed	Slower	Faster
Usage	Dynamic data	Fixed data

Why Tuple is Preferred:

- Data safety (cannot be modified)
- Faster performance
- Used in fixed data like coordinates

3. Difference between Array and List

List:

- Can store different data types
- Flexible

Array:

- Stores same data type only
- More memory efficient

Use Array When:

- Working with numeric data
- Need better performance

4. How Dictionary Works in Python

- Dictionary stores data in key-value pairs
- It uses a hash table internally

Example: `d = {"name": "Vaibhav"}`

Why Fast:

- Direct access using key
- Time complexity: $O(1)$

5. Difference between Set and List

Feature	List	Set
Duplicates	Allowed	Not Allowed
Order	Maintained	Not Maintained
Syntax	<code>[]</code>	<code>{}</code>

Example: `s = {1,2,2,3}` Output: `{1,2,3}`

Uses of Set:

- Removing duplicates
- Membership checking
- Mathematical operations

```
In [ ]: # STRING
s = "Python"
print("String:", s)
print("Indexing:", s[0])
print("Slicing:", s[1:4])
```

String: Python
Indexing: P
Slicing: yth

```
In [ ]: # LIST
lst = [1, 2, 3, 4]
lst.append(5)
print("List:", lst)
```

List: [1, 2, 3, 4, 5]

```
In [ ]: # TUPLE
tup = (10, 20, 30)
print("Tuple:", tup)
```

Tuple: (10, 20, 30)

```
In [ ]: # ARRAY
import array
arr = array.array('i', [1, 2, 3])
print("Array:", arr)
```

Array: array('i', [1, 2, 3])

```
In [ ]: # DICTIONARY
d = {"name": "Vaibhav", "age": 20}
print("Dictionary:", d)
print("Access:", d["name"])
```

Dictionary: {'name': 'Vaibhav', 'age': 20}
Access: Vaibhav

```
In [ ]: # SET
s = {1, 2, 2, 3}
print("Set:", s)
```

Set: {1, 2, 3}

```
In [ ]: # RANGE
r = range(5)
print("Range:", list(r))
```

Range: [0, 1, 2, 3, 4]

Conclusion

In this practical, we studied different sequence data types in Python and their operations like indexing and slicing. Each data type has its own importance and is useful in real-life applications.